

LAWTON FORT SILL SOUTH, OK, USA

WMO: 723575

Lat: 34.558N

Lon: 98.417W

Elev: 326

StdP: 97.47

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 03950

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-6.5	-15.8	1.0	-5.3	-12.8	1.3	-2.3	13.6	4.0	12.1	4.5	3.6	0	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	40.0	22.8	38.5	22.9	37.1	23.1	25.6	33.6	25.0	33.0	24.5	32.5	4.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.9	28.7	22.8	18.3	27.9	22.4	17.8	27.6	80.3	33.4	77.7	33.0	75.5	32.4	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	10.4	9.0	-13.0	41.3	2.7	1.7	-15.0	42.6	-16.6	43.5	-18.1	44.5	-20.0	45.7		
			-13.7	26.5	2.6	0.8	-15.5	27.1	-17.0	27.6	-18.4	28.1	-20.3	28.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	4.9	7.1	12.2	16.6	21.8	26.8	29.1	28.9	24.4	17.6	10.9	5.4
	DBStd	9.74	5.08	5.67	5.41	4.57	4.15	2.89	2.51	2.79	3.80	4.83	5.08	4.91
	HDD10.0	537	172	112	38	6	0	0	0	0	0	4	49	156
	HDD18.3	1735	418	316	199	83	16	0	0	0	3	73	226	400
	CDD10.0	3166	12	30	107	204	366	503	593	587	433	239	76	14
	CDD18.3	1322	0	1	10	32	123	253	335	328	187	49	3	0
	CDH23.3	16605	7	26	132	390	1338	3046	4594	4373	2098	541	57	3
	CDH26.7	9014	0	4	30	120	578	1628	2779	2626	1054	186	9	0
Wind	WSAvg	4.2	4.0	4.4	4.8	4.9	4.5	4.4	3.7	3.6	3.7	4.0	4.0	3.9
Precipitation	PrecAvg	738	28	33	56	64	112	97	52	63	84	72	46	37
	PrecMax	1060	82	117	173	158	415	261	214	240	190	281	155	148
	PrecMin	417	0	0	0	2	4	16	0	0	9	4	0	0
	PrecStd	189	22	26	43	43	82	63	46	56	52	68	40	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	26.5	30.0	33.1	37.3	40.0	41.9	42.2	39.8	33.5	28.0	22.8
		MCWB	12.7	15.1	16.3	18.7	21.4	22.6	22.4	22.1	21.5	20.6	17.7	13.3
	2%	DB	20.5	23.0	26.8	29.9	34.3	37.5	40.0	40.0	36.3	31.2	24.9	19.6
		MCWB	11.9	13.8	16.9	18.5	22.1	23.0	22.9	22.9	22.3	20.5	16.3	12.8
	5%	DB	17.4	20.0	24.2	27.5	32.2	35.7	38.4	38.5	34.1	28.6	22.4	17.1
		MCWB	10.3	12.4	16.1	18.4	21.6	23.1	23.2	22.9	22.2	19.1	15.2	11.1
	10%	DB	14.6	17.2	22.0	25.3	29.9	34.0	37.0	37.1	32.5	26.5	19.9	14.3
		MCWB	8.5	11.3	14.8	17.2	21.2	23.1	23.5	23.0	21.9	18.4	14.1	9.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.4	17.9	20.2	22.7	25.0	26.4	26.5	25.9	25.2	23.7	20.0	17.6
		MCDB	20.4	22.1	25.7	28.1	32.2	34.1	35.4	33.9	32.3	28.9	24.5	20.2
	2%	WB	13.5	15.9	18.8	21.2	24.0	25.5	25.5	25.2	24.2	22.5	18.0	14.6
		MCDB	17.7	20.3	24.0	26.3	30.6	33.0	33.9	33.6	31.1	27.7	22.2	17.7
	5%	WB	11.1	13.9	17.5	20.1	23.2	24.7	25.0	24.6	23.5	21.3	16.8	12.0
		MCDB	16.0	18.5	22.0	24.9	29.6	32.1	33.3	33.2	30.4	25.7	20.5	15.4
	10%	WB	9.0	11.5	15.8	19.0	22.3	24.0	24.4	24.0	22.9	20.0	15.2	9.5
		MCDB	14.5	16.9	20.7	23.5	28.1	31.2	32.9	32.7	29.7	24.5	19.4	14.2
Mean Daily Temperature Range	5% DB	MDBR	14.2	13.6	13.8	13.7	12.9	12.5	13.6	13.7	13.3	13.7	13.8	13.1
		MCDBR	20.3	19.1	17.5	17.1	16.2	15.1	16.3	16.2	15.7	16.7	17.5	18.2
		MCWBR	12.4	11.1	9.6	8.7	6.4	4.7	4.2	4.1	5.2	7.4	10.0	11.3
	5% WB	MCDBR	17.6	16.5	13.8	13.2	13.2	12.4	13.0	13.1	12.2	11.7	13.9	14.3
		MCWBR	11.8	11.3	8.4	7.9	6.4	5.1	4.3	4.1	4.5	6.2	9.3	10.6
Clear-Sky Solar Irradiance	taub	0.291	0.300	0.337	0.372	0.408	0.428	0.424	0.419	0.387	0.349	0.314	0.294	
	taud	2.518	2.507	2.421	2.349	2.318	2.301	2.347	2.304	2.370	2.462	2.497	2.508	
	Ebn at Noon	925	953	937	913	879	858	859	858	872	882	888	899	
	Edh at Noon	82	92	110	124	129	131	125	128	114	96	83	78	
All-Sky Solar Radiation	RadAvg	2.96	3.55	4.71	5.76	6.19	6.87	6.91	6.33	5.25	4.03	3.11	2.51	
	RadStd	0.30	0.50	0.30	0.46	0.66	0.60	0.41	0.36	0.42	0.56	0.39	0.28	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	-1.83	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page